Choose the Right Database for Your Project

Selecting the right database is essential for the success of any project. A good data model underpins the overall architecture, provides tremendous insights into the application itself, and builds a solid foundation for a project. There are a number of SQL (RDBMS) and NoSQL based databases available in the market. Which one should you go for?

SQL databases are based on set theory (unions and defined relationships). Here, data is stored in tables (rows and columns). These are ACID (atomicity, consistency, isolation, durability) compliant, have structured data and use SQL as the query language. They support frequent and short transactions that need to be accurate.

On the other hand, NoSQL databases are used when there are large volumes of unstructured data. These typically use cloud storage and support rapid application development. They can work with a wide variety of data models, and data is stored in a distributed fashion. They follow the shared-nothing architecture, and support BASE characteristics (basically available, soft state, eventual consistency). Unstructured data typically includes session data, click-stream data, etc.

There are different types of NoSQL databases:

- Key-value store (Redis, Memcached)
- Column store (Cassandra, HBase)
- Document (Mongo DB, CouchDB)
- Graph (Neo4j)

Evaluation questionnaire

Before deciding the right database (DB) for your application, you must query what is expected from it. You can use the sample set of questions given below, and try and be ready with the answers:

- How much volume of data is the DB going to support (GB/TB/PB, etc)?
- What is the type of data I am going to store (simple or complex data structures, etc)?
- How many concurrent users will be connecting to the database?
- What are the availability requirements for my database (e.g., 99.999%)?
- Does my database need to be highly scalable?
- What is the average response time needed for the queries?
- What is the throughput for the transactions on the database?
- Will I require a real-time consistent database or is some lag permissible?
- Does my data model undergo frequent changes?
- What are the levels of *read* and *write* operations required on the database?